Slow Relaxation of Amorphous germanium: Stretching Beyond the Stretched Exponential

Akshat Sharma* And James Kakalios

University of Minnesota, URS, Summer 2023

Abstract

The electronic transport properties of undoped hydrogenated amorphous germanium (a-Ge:H) thin films, a technologically important material, are found to be sensitive to the rate at which the film is cooled following a high temperature anneal. Two regimes are observed - a low temperature regime where the conductivity is sensitive to the cooling rate and a high temperature region where the conductivity is independent of thermal history. Following a high-temperature anneal and rapid cooling to a lower temperature (but above room temperature), the current in a-Ge:H increases as a function of time at a fixed voltage and temperature. This time dependence in amorphous semiconductors has been observed previously, with time dependencies described as a stretched exponential relaxation. However, our results indicate that a stretched exponential function is insufficient to account for current relaxation in a-Ge:H. Particularly, at 405K, conductivity follows a stretched exponential trend similar to that observed at 410K, before relatively rapidly changing to one resembling 400K after approximately 100,000 seconds of relaxation time. The mechanism underlying this jump is not understood.

Motivation/Theory

Charge Transport: The conductivity σ of amorphous semiconductor thin films as a function of temperature T can be described by an anomalous hopping expression:

$$\sigma(T) = \sigma_0 exp\left[-\left(\frac{T_0}{T}\right)^{\kappa}\right]$$

Where σ_0 is the pre-exponential factor, T_0 is a characteristic temperature and the exponent kappa (κ) indicates the type of charge transport mechanism. For band to band transport, $\kappa = 1$ while for variable range hopping, $\kappa = 0.25, 0.5, or\ 0.75$ have been observed.

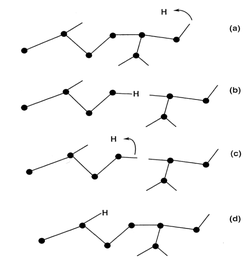
Relaxation Time Measurement: The time-dependence of σ following a rapid cooling to a temperature below the Equilibration Temperature has been studied across a temperature range of 400K to 440K. This relaxation is believed to be due to hydrogen motion in the amorphous film (top right figure) which leads to changes in the defect density of states (bottom right figure) that determines the film's conductivity. The change in σ can be normalized using the formula:

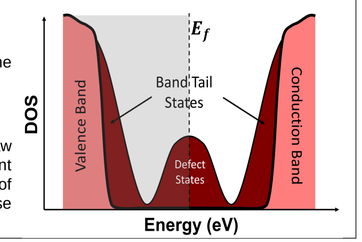
$$\Delta(t) = \frac{\sigma(t) - \sigma_f}{\sigma_i - \sigma_f}$$

Here, σ_i represent the initial conductivity, while σ_f denotes the final or saturated conductivity. The normalized value is used to express the stretched exponential

$$\Delta(T) = \exp\left[-\left(\frac{t}{\tau}\right)^{\beta}\right]$$

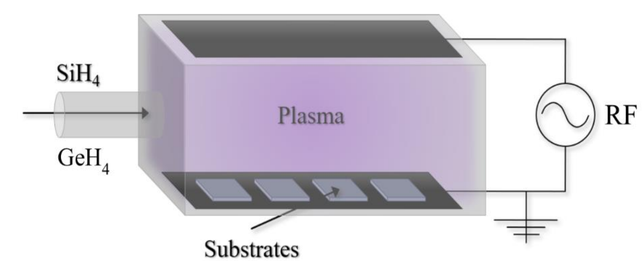
In this equation, τ signifies the relaxation lifetime and β represents the stretched exponential power-law exponent. It's worth noting that the relaxation lifetime decreases with increasing temperature. Current relaxation at temperatures near the Equilibration Temperature is highly sensitive to the specific details of the rapid cooling process, leading to greater uncertainty in the parameters β and τ obtained at these temperatures.



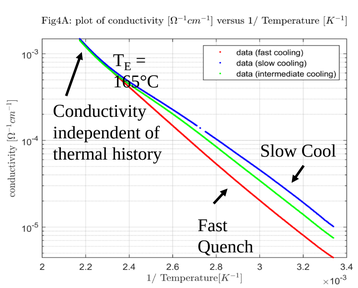


Method of Preparation

The hydrogenated amorphous Germanium (a-Ge:H) thin films studied here were synthesized using Plasma-Enhanced Chemical Vapor Deposition (PECVD) techniques. In PECVD, low-temperature plasma dissociates Germane (GeH_4), depositing a thin film of Ge, H, and dissociated GeH_4 on substrates in a vacuum chamber. For a-Ge:H, the best quality films are grown at around 250-260°C. The films studied here were 330 nm thick, and Aluminum coplanar electrodes were deposited after film growth.



Data Analysis/Results

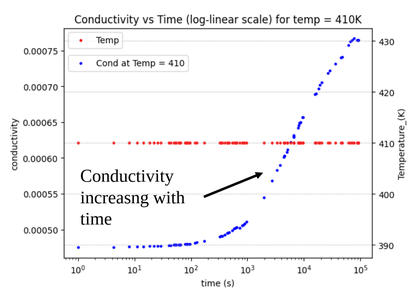


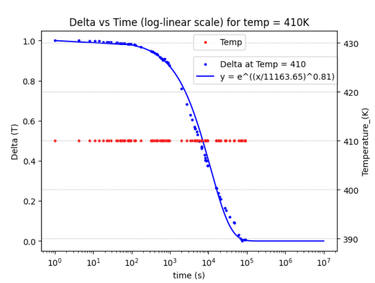
For undoped a-Ge:H, we observe two regimes:

1. 300K < T < 435K, σ sensitive to cooling rate

2. 435K < T < 490K, σ independent of thermal history

Rapid cooling freezes in a lower conductivity state than can be obtained via slow cooling

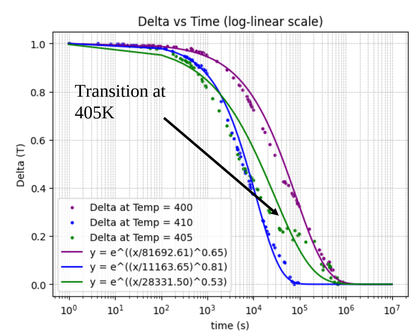


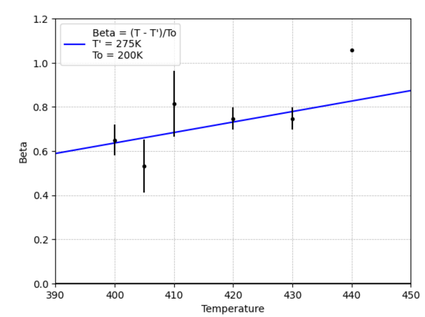


At temperature = 410K, we noticed that following a high-temperature anneal and rapid cooling to a temperature of 405K, the conductivity of the material starts to increase with time. The stretched exponential power law exponent for this temperature is close to 0.81.

In a similar manner, we performed the high temperature annealing and rapid cooling at different temperatures ranging from 400 to 440 And plotted their stretched exponential power law exponent. Until 430K, the power law exponent is approximately 0.75. However, the exponent jumps to unity at 440K and above, as the equilibrium temperature lies at 438K.

An interesting anomaly that we noticed was during relaxation at T = 405K. Initially, its conductivity follows a stretched exponential trend similar to that observed at 410K, before relatively rapidly changing to the relaxation curve measured at 400K after approximately 100,000 seconds of relaxation time. The mechanism underlying this jump is not well understood





References

- Bodurtha, K., and Kakalios, J. (2015). Non-Arrhenius anomalous hopping electronic transport in hydrogenated amorphous silicon and composite amorphous/nanocrystalline thin films. *J. of Appl. Phys.* **118**, 215103.
- *Knauber, B. (2020). "Conductance fluctuations and thermal equilibration effects in hydrogenated amorphous germanium." Ph.D. thesis dissertation, Univ. of Minnesota.*
- *Kakalios, J., Street, R. A., Jackson, W. B. (1987). "Stretched Exponential Relaxation Arising from Dispersive Diffusion of Hydrogen in Amorphous Silicon." Phys. Rev. Lett.* ***59****, 1037.*